

[58]**SARDAR PATEL UNIVERSITY****BCA Semester-5 External Examination, October - 2025**
US05CBCA23 || SOFTWARE ENGINEERING [Old Course]**Date : 07-10-2025****Time : 10.00 a.m. – 01.00 p.m.****Total : 70 Marks****Q-1 Select the appropriate option for the following questions:****10**

- 1) _____ is the second step of design phase.
A. Design Analysis C. Black box
B. System Design D. Detailed Design
- 2) _____ model overcomes first two limitations of Waterfall model.
A. Spiral C. Iterative enhancement
B. Prototype D. Waterfall
- 3) A high quality SRS reduces the development _____.
A. Time C. Cost
B. Customer requirements D. Quality
- 4) An SRS provides a reference for _____ of the final product.
A. Validation C. Verification
B. Xerox copy D. Quality
- 5) Which one is the key term used in design of a system?
A. Module C. Process
B. Data D. None
- 6) In functional abstraction the module considered as _____ for detail design.
A. White box C. Compiled box
B. Black box D. None
- 7) Which is not the level of Cohesion?
A. Logical C. Sequential
B. Physical D. Coincidental
- 8) In programming style, nesting means _____.
A. switch C. nested function
B. if-then-else D. nested for loop
- 9) When _____ type of variables is changed then some side effects are occurred.
A. static C. global
B. dynamic D. None of these
- 10) _____ activity is used to understand the needs, goals and constraints.
A. Testing C. Design
B. Requirement Specification D. Problem Analysis

Q-2 Give short answers of the following questions: (Attempt TEN out of TWELVE) 20

- 1) List characteristics of software process.
- 2) Write a short note on Requirement analysis phase.
- 3) Draw diagram of iterative enhancement model.
- 4) What is Structured English?
- 5) Explain Projections.

[P.T.O.]**(1)**

- 6) Explain Partitioning.
- 7) List the levels of Cohesion.
- 8) Explain in brief Design walkthrough.
- 9) Define: Module and Modular System.
- 10) Write at least 2 differences between functional and structural testing.
- 11) What is a goal of coding?
- 12) List any four automated testing tools.

Q-3 A Explain testing phase. **05**

B Explain Design phase. **05**

OR

Q-3 A Explain prototype model. **05**

B Explain Waterfall model. **05**

Q-4 A Explain general characteristics of SRS. **06**

B What is SRS? Explain needs of SRS. **04**

OR

Q-4 A Explain COCOMO Model. **06**

B Explain validation process of SRS. **04**

Q-5 A Write brief note on Structured design. **06**

B Write a short note on Coupling. **04**

OR

Q-5 A Discuss the basic principles of design in detail. **06**

B Explain verification techniques for system design. **04**

Q-6 What is testing? Explain levels of testing in detail with diagrams. **10**

OR

Q-6 Explain programming style and coding approach. **10**

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